

Annex A

Foundations of Reticular Local Abstraction and Compact Reticular Computability (with Hook to Epistemic CNNs)

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Abstract

This annex provides the foundational definitions, axioms, and basic results for *Reticular Local Abstraction* (RLA) and *Compact Reticular Computability* (CRC), as used throughout the main position paper and technical annexes B–E. The focus is strictly epistemic: RLA and CRC are presented as tools for reasoning about the structure and limits of *multi-level models* and epistemic architectures, not as ontological claims about the world.

An RLA system is a finite reticulum of levels of abstraction $\{L_i\}$, each with its own state space and rule set, connected by transmissions $\tau_{i \rightarrow j}$ that may be injective, quasi-injective, or non-injective (information collapse). CRC characterises when such a reticulum is: (i) internally ontologically autonomous (relative to a modelling horizon), (ii) epistemically closed, and (iii) fully Turing-computable. Under these conditions, undecidability and emergence become structural properties of the *modelling reticulum*: undecidable properties at Turing-like levels can propagate along quasi-injective transmissions, while non-injective collapses can generate macro properties that are not definable at lower levels, yet remain computable within their own domain.

After introducing the core notions, we state and sketch a key theorem on propagation of undecidability, and define operational metrics such as the *coefficient of collapse*, an *index of emergence*, and a *reticular irreducibility index*. A concrete toy example with three finite levels and explicit numerical values is provided to illustrate how these indices are actually computed in practice. We then provide a structural mapping from convolutional neural networks (CNNs) to RLA reticula and outline how *Epistemic CNNs* (ECNNs) instantiate CRC in engineered architectures. Technical details and domain-specific developments are deferred to: Annex B (compact RLA topology and CRC for a generalist bryophyte), Annex C (ECNN formalisation and training), Annex D (RLA–ECNN bridge to the Principle of Computational Equivalence and dynamical diagnostics), and Annex E (Epistemic Computational Units / Entities and protocols).

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1 Introduction and Epistemic Stance

1.1 Multi-level models and classical computability

Classical computability theory, from Turing’s analysis of effective procedures [Turing, 1936/1937] and Church’s λ -calculus [Church, 1936] to Rice’s theorem on extensional properties of programs [Rice, 1953], provides a complete account of which functions on discrete domains are computable. In that framework, systems are typically represented by *single-level* formalisms: Turing machines, register machines, or other equivalent models. Questions of decidability and undecidability are posed within this single configuration space.

By contrast, most scientific practice employs *multi-level* modelling. Systems biology connects molecular, cellular, tissue and organismal dynamics [Noble, 2012]; condensed matter physics moves between microscopic configurations and emergent phases of matter [Laughlin & Pines, 2000]; cognitive science and neuroscience relate neuronal circuits, network dynamics, and behavioural or

cognitive states [Bechtel & Abrahamsen, 2005]. These modelling chains involve not only multiple formalisms but also systematic use of coarse-graining, aggregation, and information collapse.

Classical computability is agnostic about this multi-level structure. It tells us what is computable at a given level, but not how computability and undecidability behave across chains of levels linked by information-losing or information-preserving transformations. In the perspective developed here, classical single-level machines appear as degenerate RLA reticula with one level and no transmissions; CRC leaves all standard results of computability theory untouched and simply extends the analysis to explicitly multi-level architectures.

1.2 RLA and CRC as epistemic, not ontological, frameworks

The RLA/CRC framework introduced here is explicitly *epistemic*. Levels, transmissions, and reticula are not presented as literal layers of an objective ontology; they are abstractions of how scientific communities carve up, represent, and reason about their domains.

When we say that a level is “Turing-like”, we mean that the *formalism* used to model that slice of practice can, in principle, emulate universal computation. When we speak of an “internal ontological horizon”, we refer to the ontology induced by a modelling framework—the entities and laws it deems explanatory—rather than to an ultimate ontology of the world. Nothing in this annex presupposes a particular metaphysics of nature; the focus is on the structure of models.

Within this stance:

- undecidability is understood as a property of the formalisms used at certain levels;
- emergence is treated as a property of the way lower-level models are collapsed into higher-level ones;
- computability is extended from single machines to reticula of models.

The goal of Annex A is to define these ideas precisely enough to support the technical constructions and case studies in Annexes B–E.

2 Core Constructs of Reticular Local Abstraction

2.1 Levels of abstraction as epistemic slices

Definition 2.1 (Level of abstraction). *A level of abstraction (or simply level) is a pair*

$$L = \langle D(L), \Sigma(L) \rangle,$$

where $D(L)$ is a set (or discretizable space) of admissible states and $\Sigma(L)$ is a set of rules that act on $D(L)$. Rules may represent dynamical laws, update equations, inference rules, or any effective procedures used at that level in the relevant scientific practice.

Examples include: ordinary differential equations governing population densities, Markov models for gene expression, state machines for control logic, or neural network modules defined over feature spaces.

2.2 Transmissions between levels

Definition 2.2 (Transmission). *Given two levels $L_i = \langle D(L_i), \Sigma(L_i) \rangle$ and $L_j = \langle D(L_j), \Sigma(L_j) \rangle$, a transmission from L_i to L_j is a mapping*

$$\tau_{i \rightarrow j} : D(L_i) \rightarrow \mathcal{P}(D(L_j))$$

from states at L_i to sets of possible states at L_j . In many applications we restrict to single-valued maps $\tau_{i \rightarrow j} : D(L_i) \rightarrow D(L_j)$ for simplicity.

Transmissions formalise how states at one level are translated, coarse-grained, or lifted to another level. They can be:

- injective on a domain of interest,
- quasi-injective on a critical subset,
- or genuinely non-injective (many-to-one) on relevant states.

Definition 2.3 (Injective, quasi-injective, non-injective). *Let $S \subseteq D(L_i)$ be a subset of interest. A transmission $\tau_{i \rightarrow j}$ is:*

- injective on S if $x_1, x_2 \in S$ and $\tau_{i \rightarrow j}(x_1) = \tau_{i \rightarrow j}(x_2)$ imply $x_1 = x_2$;
- quasi-injective on S if there exists a partition $S = S' \cup S''$ such that $\tau_{i \rightarrow j}$ is injective on S' and S'' is declared irrelevant for a given analysis (e.g. below resolution);
- non-injective on S if there exist distinct $x_1, x_2 \in S$ with $\tau_{i \rightarrow j}(x_1) = \tau_{i \rightarrow j}(x_2)$ and the distinction between x_1, x_2 is relevant for at least one property of interest.

Non-injective transmissions implement *information collapse*: coarse-graining, saturations, thresholds, or aggregation moves that merge distinct lower-level states into one macro-state.

2.3 RLA reticula and Turing-like levels

Definition 2.4 (RLA reticulum). *A Reticular Local Abstraction (RLA) reticulum is a finite sequence*

$$R = (L_1, \dots, L_n)$$

of levels of abstraction together with transmissions $\tau_{i \rightarrow i+1} : D(L_i) \rightarrow D(L_{i+1})$ for $i = 1, \dots, n-1$. We sometimes write $\mathcal{L}(R) = \{L_1, \dots, L_n\}$ and $\mathcal{T}(R) = \{\tau_{i \rightarrow i+1}\}$.

Definition 2.5 (Turing-like level). *A level $L_i = \langle D(L_i), \Sigma(L_i) \rangle$ is Turing-like if the formalism represented by $\Sigma(L_i)$ can, up to a computable encoding of inputs and outputs, simulate a universal Turing machine (or an equivalent model of computation).*

The Turing-like label attaches to the *formalism* at the level, not to the underlying physical system. For instance, a system of ODEs may be Turing-like if it encodes universal computation under a suitable interpretation; similarly, a sufficiently flexible programming language or neural net may be Turing-like.

3 Compact Reticula and Compact Reticular Computability

3.1 Epistemic horizons

Definition 3.1 (Epistemic horizon). *An epistemic horizon $\mathcal{H}$ for a modelling practice is a specified set of observables, questions, and explanatory tasks that the modeller is prepared to treat as in-scope for a given reticulum R . Formally, $\mathcal{H}$ is a finite or countable family of predicates and queries about the domain, together with a specification of which answers or explanations are considered admissible.*

The epistemic horizon is part of the modelling choice: different communities may choose different horizons for the same underlying system, yielding different reticula and different judgements of compactness.

3.2 Compact reticula: IOA, EC, and TC

To treat an RLA reticulum as a well-posed computational object, we require three conditions, understood epistemically.

Definition 3.2 (Internal Ontological Autonomy (IOA)). *An RLA reticulum R satisfies internal ontological autonomy with respect to a domain of discourse if all explanatory entities and laws invoked for that domain are represented internally by:*

- *state spaces $D(L_i)$ and rule sets $\Sigma(L_i)$, and*
- *transmissions $\tau_{i \rightarrow j}$ between levels.*

External inputs (e.g. environmental drivers) may appear as boundary conditions or exogenous parameters but are not modelled as additional explanatory entities outside $\mathcal{L}(R)$.

Definition 3.3 (Epistemic Closure (EC)). *An RLA reticulum R is epistemically closed with respect to an epistemic horizon $\mathcal{H}$ if, for each observable or question in $\mathcal{H}$, there exists a level (or finite subset of levels) whose state spaces and rules suffice to represent and explain it internally, using only $\mathcal{L}(R)$ and $\mathcal{T}(R)$.*

Definition 3.4 (Turing-Computability (TC)). *An RLA reticulum R satisfies Turing-computability if:*

- *for each L_i , the rule set $\Sigma(L_i)$ is effectively computable by some Turing machine, and*
- *for each transmission $\tau_{i \rightarrow j}$, the mapping is algorithmically implementable (up to any required numerical precision).*

Definition 3.5 (Compact reticulum). *An RLA reticulum R is compact (in the epistemic sense) if it satisfies IOA, EC, and TC with respect to a specified epistemic horizon $\mathcal{H}$.*

Compactness is therefore relative to the chosen horizon: a reticulum may be compact for one family of questions and not for another.

3.3 Compact Reticular Computability (CRC)

Definition 3.6 (Compact Reticular Computability (CRC)). *A compact reticulum R is compactly reticularly computable if:*

1. *there exists a Turing machine T that simulates every $\Sigma(L_i)$ and $\tau_{i \rightarrow j}$ to arbitrary precision;*
2. *there exists at least one level L_k that is Turing-like;*
3. *there exist macro-properties at higher levels L_j , $j > k$, that are computable with respect to $\Sigma(L_j)$ but not definable, with respect to the formal language and rule sets used at those levels, as predicates on $D(L_i)$ for $i < j$.*

CRC is thus a property of multi-level *models*: it says that the entire reticulum behaves as a single computational object with internal logical limits and emergent properties. Classical computability is recovered as the degenerate case where R has a single level and no transmissions.

3.4 Critical sets and propagation of undecidability

We now formalise one key mechanism: how undecidability at a Turing-like level behaves under quasi-injective transmissions.

Definition 3.7 (Critical set). *Let $L_i = \langle D(L_i), \Sigma(L_i) \rangle$ be a Turing-like level. A subset $C_i \subseteq D(L_i)$ is critical if the restriction of $\Sigma(L_i)$ to C_i is Turing-complete: for any Turing machine M and input w , there exists $x \in C_i$ such that the behaviour of $\Sigma(L_i)$ on x encodes the behaviour of M on w , up to a computable coding.*

Definition 3.8 (Extensional property). *Let L_i and C_i be as above. A subset $U \subseteq C_i$ is an extensional property (in the sense of Rice) if membership of x in U depends only on the observable behaviour of x under $\Sigma(L_i)$: whenever $x_1, x_2 \in C_i$ have identical behaviour under $\Sigma(L_i)$, either both belong to U or both are outside U .*

Axiom 3.1 (Undecidability at Turing-like levels). *Let L_i be Turing-like and C_i be a critical set. Then any non-trivial extensional property $U \subseteq C_i$ is undecidable with respect to $\Sigma(L_i)$, in the sense of Rice's theorem [Rice, 1953].*

Theorem 3.1 (Propagation of undecidability along quasi-injective transmissions). *Let R be a compact reticulum and L_i be a Turing-like level with critical set C_i . Suppose $\tau_{i \rightarrow i+1}$ is effectively computable and injective on C_i . Then for any non-trivial extensional property $U \subseteq C_i$ that is undecidable with respect to $\Sigma(L_i)$, the induced property*

$$U' := \{y \in \tau_{i \rightarrow i+1}(C_i) \mid \exists x \in C_i : y = \tau_{i \rightarrow i+1}(x), x \in U\}$$

is undecidable with respect to the induced dynamics on $\tau_{i \rightarrow i+1}(C_i)$.

Sketch. If U' were decidable at the higher level, we could decide U at L_i by: (i) mapping x to $y = \tau_{i \rightarrow i+1}(x)$, (ii) deciding whether $y \in U'$, and (iii) using injectivity on C_i to infer $x \in U$ iff $y \in U'$. This would contradict Axiom 3.1. The argument is a standard Rice-style reduction. $\square$

A similar statement holds for quasi-injective transmissions on a restricted domain $C'_i \subseteq C_i$ where undecidable properties of interest are defined. The main message is that undecidability is not automatically “washed out” by going up a modelling hierarchy; it persists whenever relevant distinctions are preserved.

3.5 Emergence via non-injective collapse

Non-injective transmissions, by contrast, implement deliberate information loss. They can block the propagation of certain undecidable properties and, at the same time, generate new macro-properties not definable at lower levels.

Axiom 3.2 (Emergence via collapse). *Let $\tau_{i \rightarrow j} : D(L_i) \rightarrow D(L_j)$ be non-injective on a subset $S \subseteq D(L_i)$ that is relevant to property families of interest. Then, given the descriptive resources of the modelling language at L_i , there exists at least one macro-property Q of states in $D(L_j)$ that is not definable as a predicate on $D(L_i)$ alone.*

Proposition 3.1 (Emergent macro-properties). *Under Axiom 3.2, at least one macro-property Q at L_j qualifies as emergent with respect to L_i : it is computable at L_j but not reducible to a predicate on $D(L_i)$ without supplementing the language or rules.*

Sketch. Intuitively, non-injective collapse identifies lower-level states that differ in ways that matter for some candidate properties. Any property that depends only on the equivalence classes induced by $\tau_{i \rightarrow j}$ can be defined at L_j without reference to the collapsed distinctions. At least one such property will not be definable without quotienting by $\tau_{i \rightarrow j}$, given reasonable assumptions on the language. A detailed treatment is given in the extended technical development accompanying this annex. $\square$

Together, Theorem 3.1 and Axiom 3.2 capture the coexistence of logical limits and emergent structure in compact reticula.

4 Operational Diagnostics and Metrics

The following diagnostics are not axioms but *operational proxies* for reticular phenomena. They are intended for empirical and numerical studies of specific reticula, including the engineered architectures of Annex C.

4.1 Coefficient of collapse

Definition 4.1 (Coefficient of collapse). *Let $\tau_{i \rightarrow j} : D(L_i) \rightarrow D(L_j)$ be a transmission and $S \subseteq D(L_i)$ a finite reference set. The coefficient of collapse on S is*

$$CC_{i \rightarrow j}(S) := 1 - \frac{|\tau_{i \rightarrow j}(S)|}{|S|} \in [0, 1],$$

where $\tau_{i \rightarrow j}(S)$ denotes the image set (up to a chosen numerical tolerance).

$CC_{i \rightarrow j}(S) \approx 0$ indicates near-injective behaviour on S ; $CC_{i \rightarrow j}(S) \approx 1$ indicates strong many-to-one collapse. In Annex C this diagnostic is specialised to ECNN architectures as $CC_{i \rightarrow j}^{(\text{ECNN})}(S)$, computed on activation tensors.

4.2 Index of emergence

Definition 4.2 (Index of emergence). *Let $\mathcal{Q}$ be a finite family of candidate macro-properties defined over $D(L_j)$. For a given pair of levels (L_i, L_j) , the index of emergence is defined as*

$$IE_{i \rightarrow j} := \frac{\#\{Q \in \mathcal{Q} : Q \text{ is computable at } L_j \text{ and not definable on } D(L_i)\}}{|\mathcal{Q}|} \in [0, 1].$$

$IE_{i \rightarrow j}$ quantifies, within the chosen family of candidate properties, how many are genuinely emergent with respect to L_i . In practice, definability tests will rely on the expressive resources of the modelling language at L_i and on approximations (e.g. linear probes, symbolic regression). Annex C introduces $IE_{i \rightarrow j}^{(\text{ECNN})}$ as a concrete instantiation for ECNNs.

4.3 Reticular irreducibility index

Inspired by computational mechanics [Shalizi & Crutchfield, 2001], we introduce a structural index for reticula.

Definition 4.3 (Reticular irreducibility index). *Consider a causal graph G_t induced by levels and transmissions in a reticulum R over a finite horizon t , with vertex set $V(G_t)$. Let $\text{MinCut}(G_t)$ be the size of a minimum vertex cut separating inputs from outputs over that horizon. The reticular irreducibility index over a time horizon T is*

$$RIR(R; T) := 1 - \frac{1}{T} \sum_{t=1}^T \frac{\text{MinCut}(G_t)}{|V(G_t)|} \in [0, 1].$$

Low RIR indicates that small cuts suffice to summarise dynamics, suggesting high compressibility. High RIR indicates that no small cut suffices, suggesting computational irreducibility in the sense of [Shalizi & Crutchfield, 2001, Wolfram, 2002]. In Annex D, RIR is used in conjunction with cellular automata mappings to connect CRC reticula to dynamical regimes in the Principle of Computational Equivalence (PCE). Annex C outlines network-specific estimators of RIR for ECNNs.

4.4 Toy numerical example: three-level finite reticulum

To make the above diagnostics more concrete, we present a small toy RLA reticulum with three finite levels and explicit numerical values for CC , IE , and RIR . The example is intentionally simple and pedagogical rather than realistic.

Structure of the toy reticulum

Consider an RLA reticulum

$$R_{\text{toy}} = (L_1, L_2, L_3)$$

with three levels and finite state spaces:

- Micro-level L_1 with

$$D(L_1) = \{s_1, s_2, s_3, s_4\};$$

- Meso-level L_2 with

$$D(L_2) = \{u_1, u_2, u_3\};$$

- Macro-level L_3 with

$$D(L_3) = \{v_1, v_2\}.$$

Transmissions are defined as follows. The first transmission $\tau_{1 \rightarrow 2} : D(L_1) \rightarrow D(L_2)$ is

$$\tau_{1 \rightarrow 2} : \begin{cases} s_1 \mapsto u_1, \\ s_2 \mapsto u_1, \\ s_3 \mapsto u_2, \\ s_4 \mapsto u_3, \end{cases}$$

and the second transmission $\tau_{2 \rightarrow 3} : D(L_2) \rightarrow D(L_3)$ is

$$\tau_{2 \rightarrow 3} : \begin{cases} u_1 \mapsto v_1, \\ u_2 \mapsto v_1, \\ u_3 \mapsto v_2. \end{cases}$$

The composite transmission $\tau_{1 \rightarrow 3} := \tau_{2 \rightarrow 3} \circ \tau_{1 \rightarrow 2}$ thus maps

$$s_1, s_2, s_3 \mapsto v_1, \quad s_4 \mapsto v_2.$$

Table 1 summarises the structure of R_{toy} .

Table 1: Toy three-level reticulum R_{toy} : states and images under transmissions.

Level	States	Transmission images
L_1	s_1	$\tau_{1 \rightarrow 2}(s_1) = u_1, \tau_{1 \rightarrow 3}(s_1) = v_1$
	s_2	$\tau_{1 \rightarrow 2}(s_2) = u_1, \tau_{1 \rightarrow 3}(s_2) = v_1$
	s_3	$\tau_{1 \rightarrow 2}(s_3) = u_2, \tau_{1 \rightarrow 3}(s_3) = v_1$
	s_4	$\tau_{1 \rightarrow 2}(s_4) = u_3, \tau_{1 \rightarrow 3}(s_4) = v_2$
L_2	u_1	$\tau_{2 \rightarrow 3}(u_1) = v_1$
	u_2	$\tau_{2 \rightarrow 3}(u_2) = v_1$
	u_3	$\tau_{2 \rightarrow 3}(u_3) = v_2$
L_3	v_1	—
	v_2	—

Coefficient of collapse

Let $S_1 = D(L_1) = \{s_1, s_2, s_3, s_4\}$. The image of S_1 under $\tau_{1 \rightarrow 2}$ is

$$\tau_{1 \rightarrow 2}(S_1) = \{u_1, u_2, u_3\},$$

so $|\tau_{1 \rightarrow 2}(S_1)| = 3$ and $|S_1| = 4$. By definition,

$$CC_{1 \rightarrow 2}(S_1) = 1 - \frac{|\tau_{1 \rightarrow 2}(S_1)|}{|S_1|} = 1 - \frac{3}{4} = \frac{1}{4} = 0.25.$$

Thus $\tau_{1 \rightarrow 2}$ identifies a single pair of micro-states (s_1, s_2) , producing a mild collapse.

For the second transmission we take $S_2 = \tau_{1 \rightarrow 2}(S_1) = \{u_1, u_2, u_3\}$. Its image is

$$\tau_{2 \rightarrow 3}(S_2) = \{v_1, v_2\},$$

so $|\tau_{2 \rightarrow 3}(S_2)| = 2$ and $|S_2| = 3$, and

$$CC_{2 \rightarrow 3}(S_2) = 1 - \frac{|\tau_{2 \rightarrow 3}(S_2)|}{|S_2|} = 1 - \frac{2}{3} = \frac{1}{3} \approx 0.33.$$

Collapse is already present between L_1 and L_2 and becomes slightly stronger between L_2 and L_3 .

Index of emergence

We now introduce a small family $\mathcal{Q}$ of candidate macro-properties on $D(L_3)$ and classify them as emergent or non-emergent with respect to L_1 .

Let $\mathcal{Q} = \{Q_1, Q_2, Q_3, Q_4\}$, where:

- Q_1 : “the macro-state is v_1 ”;
- Q_2 : “the macro-state is v_2 ”;
- Q_3 : “the macro-state corresponds (under $\tau_{1 \rightarrow 3}$) to more than one micro-state”;
- Q_4 : “the macro-state corresponds to at least three micro-states under $\tau_{1 \rightarrow 3}$ ”.

In this toy,

$$\tau_{1 \rightarrow 3}^{-1}(v_1) = \{s_1, s_2, s_3\}, \quad \tau_{1 \rightarrow 3}^{-1}(v_2) = \{s_4\}.$$

We assume that the modelling language at L_1 can refer to individual micro-states in $D(L_1)$ but cannot explicitly quantify the *cardinality* of their preimages under $\tau_{1 \rightarrow 3}$ (i.e. it cannot talk about “how many” micros collapse onto a given macro).

Under this assumption:

- Q_1 and Q_2 are non-emergent: they simply select one of the two macro-states and can be defined at L_1 as the subsets $\{s_1, s_2, s_3\}$ and $\{s_4\}$, respectively.
- Q_3 and Q_4 are emergent: they depend on the multiplicity of the preimage of a macro-state under $\tau_{1 \rightarrow 3}$ and cannot be expressed, by hypothesis, as predicates on $D(L_1)$ that ignore the collapse structure.

We therefore classify

$$Q_1, Q_2 \text{ as non-emergent,} \quad Q_3, Q_4 \text{ as emergent,}$$

so that the empirical index of emergence for this toy reticulum is

$$IE_{1 \rightarrow 3} = \frac{\#\{\text{emergent } Q \in \mathcal{Q}\}}{|\mathcal{Q}|} = \frac{2}{4} = 0.5.$$

Reticular irreducibility index

We now construct a simple causal graph G_1 for the toy reticulum over a single time step and compute RIR .

For $t = 1$, let G_1 be the directed acyclic graph with vertex set

$$V(G_1) = D(L_1) \cup D(L_2) \cup D(L_3) = \{s_1, s_2, s_3, s_4, u_1, u_2, u_3, v_1, v_2\},$$

so $|V(G_1)| = 9$, and with edges

- from each s_i to $\tau_{1 \rightarrow 2}(s_i)$,
- from each u_k to $\tau_{2 \rightarrow 3}(u_k)$.

We are interested in separating the micro-level nodes $D(L_1)$ from the macro-level nodes $D(L_3)$ via a vertex cut. In this three-layer DAG a natural minimal vertex cut is given by removing all nodes at level L_2 , i.e. $\{u_1, u_2, u_3\}$: removing fewer than three vertices leaves at least one directed path from some s_i to some v_j .

Thus we may take

$$\text{MinCut}(G_1) = 3, \quad |V(G_1)| = 9.$$

Restricting attention to a single time horizon $T = 1$, the reticular irreducibility index is

$$RIR(R_{\text{toy}}; 1) = 1 - \frac{\text{MinCut}(G_1)}{|V(G_1)|} = 1 - \frac{3}{9} = \frac{2}{3} \approx 0.67.$$

Even in this minimal example, a non-negligible fraction of the reticular structure must be retained to preserve the mapping from micro-states to macro-states: the intermediate level cannot be collapsed to a single effective node without loss.

Summary

Table 2 summarises the values of the three diagnostics in this toy case.

This example is deliberately minimal, but it makes explicit how one can compute CC , IE , and RIR by hand for a finite RLA reticulum. In realistic applications (e.g. the bryophyte case of Annex B or ECNNs in Annex C), the same definitions apply, with indices estimated numerically over sampled state sets and time horizons.

Table 2: Toy numerical values of collapse, emergence, and reticular irreducibility for the three-level reticulum R_{toy} .

Index	Definition in toy case	Value
$CC_{1 \rightarrow 2}$	$1 - \tau_{1 \rightarrow 2}(S_1) / S_1 $ with $S_1 = D(L_1)$	0.25
$CC_{2 \rightarrow 3}$	$1 - \tau_{2 \rightarrow 3}(S_2) / S_2 $ with $S_2 = \tau_{1 \rightarrow 2}(D(L_1))$	≈ 0.33
$IE_{1 \rightarrow 3}$	fraction of emergent $Q \in \{Q_1, Q_2, Q_3, Q_4\}$ wrt L_1	0.5
$RIR(R_{\text{toy}}; 1)$	$1 - \text{MinCut}(G_1)/ V(G_1) $	≈ 0.67

5 From RLA to Neural Architectures

5.1 CNNs as RLA reticula

Convolutional neural networks (CNNs) [LeCun et al., 2015, Goodfellow et al., 2016] exhibit a multi-level architecture that maps naturally onto the RLA framework. We view their layers as levels and their interconnections as transmissions.

Consider a standard CNN with:

- input layer L_0 (e.g. images),
- convolutional layers $L_1, \dots, L_k$,
- pooling layers interleaved among them,
- dense (fully connected) layers $L_{k+1}, \dots, L_m$,
- output layer L_{m+1} producing logits or class probabilities.

We define:

$$L_i = \langle D(L_i), \Sigma(L_i) \rangle,$$

where $D(L_i)$ is the space of activation tensors at layer i and $\Sigma(L_i)$ consists of the forward computation rules (linear transforms plus non-linearities) and, if applicable, recurrent or skip connections.

Transmissions $\tau_{i \rightarrow i+1}$ are implemented by the layer transformations:

- convolutional layers are often approximately quasi-injective on discriminative patterns (up to symmetries and numerical approximation);
- pooling layers (max or average pooling) are non-injective: they implement explicit collapses over local neighbourhoods, generating invariances to small transformations at the cost of information loss;
- dense layers are typically many-to-one mappings from rich feature spaces to compact representations or logits.

Under this reading, a CNN is a particular RLA reticulum where non-injective transmissions play a central role in generalisation and invariance creation.

5.2 Compactness and CRC in CNN reticula

Proposition 5.1 (Compactness of trained CNN reticula). *Let R_{CNN} be the RLA reticulum induced by a trained CNN for a task with epistemic horizon $\mathcal{H}$ consisting of: inputs, labels, and a fixed family of explanatory questions (e.g. class membership, confidence, feature attributions). If:*

- *all computations in the network are implementable by a Turing machine with finite precision (TC);*
- *all explanatory entities and laws for $\mathcal{H}$ are internal to the network’s layers and transmissions (IOA);*
- *every question in $\mathcal{H}$ can be answered or represented using only the network’s states and rules (EC);*

then R_{CNN} is a compact reticulum with respect to $\mathcal{H}$.

Sketch. TC follows from the finite, algorithmic nature of forward passes and any recurrent computations. IOA holds by hypothesis, since all explanatory entities (features, weights, activations) are internal to the network. EC holds because, by assumption, all questions in $\mathcal{H}$ can be expressed and answered using only network states and rules. Thus the definition of compact reticulum is satisfied. $\square$

Proposition 5.2 (CRC in CNN reticula). *If, in addition, some level of R_{CNN} is Turing-like and there exist macro-properties at higher levels that are computable there but not definable as predicates on earlier levels (relative to their modelling languages), then R_{CNN} is an instance of Compact Reticular Computability (CRC).*

Sketch. The conditions of the CRC definition are satisfied: a global Turing machine can simulate all levels; a Turing-like level exists; and emergent macro-properties at higher levels are not definable on earlier levels. Hence R_{CNN} is CRC. $\square$

In this reading, the non-injective pooling and dense layers of CNNs provide explicit examples of collapse-induced emergence in engineered systems, while Turing-like substructures (e.g. recurrent cores or architectures with sufficient memory) provide loci for undecidability in the sense of Rice’s theorem.

6 Hook to Epistemic CNNs (ECNNs)

6.1 Limitations of standard CNN heads

Standard CNNs are typically trained with softmax outputs and cross-entropy loss. Their limitations with respect to epistemic behaviour are well known:

- They often exhibit overconfident predictions and poor calibration [Guo et al., 2017].
- They lack explicit mechanisms to output “I do not know” or “this input contradicts my training distribution”.

- Out-of-distribution (OOD) samples can elicit high-confidence but incorrect outputs [Hendrycks & Gimpel, 2016].

In RLA terms, the final transmission to the output layer is not designed to make epistemic limits visible as first-class objects. Undecidability, ignorance, and contradiction remain implicit, if they appear at all.

6.2 Epistemic heads and ECNNs as CRC instances

Epistemic CNNs (ECNNs), developed in detail in Annex C, augment CNN architectures with an *epistemic channel* and an *epistemic head* that produce:

- standard predictions or distributions over labels;
- explicit “unknown” and “contradiction” states;
- additional artefacts such as confidence scores, rationales, and flags for OOD or adversarial inputs.

The epistemic head is trained under a composite loss that balances prediction accuracy with:

- calibration (e.g. via Brier score or expected calibration error),
- robust abstention under uncertainty (e.g. following optimal reject-option theory [Chow, 1970]),
- and detection of internal inconsistencies and challenge-set contradictions.

In the RLA/CRC framework, an ECNN becomes an *engineered CRC instance*:

- its levels and transmissions define a compact reticulum;
- its epistemic head makes ignorance and contradiction into computable observables at the highest level;
- its behaviour can be evaluated against Popper-style challenge suites (Popper- χ) that test epistemic and predictive performance (Annex C).

Annex C introduces ECNN-specific versions of *CC*, *IE*, and *RIR*, and shows how to measure emergence and falsifiability in concrete architectures. Annex D then connects these diagnostics to dynamical regimes in the Principle of Computational Equivalence.

6.3 Connection to ECU/UCE architectures

The epistemic head of an ECNN can be realised with varying degrees of complexity. At one extreme, it may be a small feed-forward network that maps final features to epistemic states. At the other extreme, it may be implemented as a composition of *Epistemic Computational Units* (ECUs) and *Epistemic Computational Entities* (UCEs), described in Annex E:

- An ECU is a bounded large language model instance operating under a deterministic epistemic matrix M , capable of generating structured epistemic artefacts from feature representations.

- UCEs are topologically organised rings of ECUs that can implement executive, valuation, memory, and metacognitive functions in an epistemically closed, Turing-computable way.

From the perspective of Annex A, this shows that:

- the highest levels of a CRC reticulum need not be limited to scalar activations;
- they can instead encode high-level epistemic objects, as long as these admit computable representations;
- the crucial structural features are the reticular organisation, the injective vs non-injective transmissions, and the satisfaction of IOA, EC, and TC.

In this sense, Annex A provides the general language in which ECNNs, ECUs, and UCEs can be treated as specific, engineered instances of reticular, compact, and epistemically expressive computational systems.

7 Summary and Relation to Annexes B–E

Annex A has defined:

- RLA reticula as chains of levels linked by transmissions;
- compact reticula via IOA, EC, and TC, relative to an epistemic horizon;
- CRC as the extension of computability to such reticula;
- axioms and basic results on undecidability and emergence;
- operational indices *CC*, *IE*, and *RIR*, illustrated by a fully worked toy example;
- and a structural mapping from CNNs to RLA reticula, with an explicit hook to ECNNs.

The subsequent annexes build on this foundation:

- Annex B instantiates RLA and CRC in a natural domain: a compact reticular lattice for a generalist bryophyte, built from plant physiology and ecology, providing an empirical reference case.
- Annex C develops ECNNs as engineered CRC instances with epistemic heads, reticular diagnostics, and Popper- χ evaluation protocols.
- Annex D (RLA-ECNN bridge to PCE) leverages *CC*, *IE*, and *RIR* to connect CRC reticula to cellular automata and to discuss dynamical regimes in the sense of the Principle of Computational Equivalence.
- Annex E specifies ECU/UCE architectures as implementations of epistemic heads and higher-level reticula where the basic “neurons” are LLM-based epistemic units operating over high-level objects constrained by epistemic matrices.

Together, these annexes provide a layered technical background for the main position paper. Annex A should be read as the minimal, self-contained foundation on which the other, more specialised constructions rest.

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